MindCapsule — Hug the Hedgehog
Development Process Documentation

1. Project Theme and Objectives

This project focuses on high school students who experience anxiety and academic stress but may hesitate to seek help due to emotional stigma. The system aims to create a private and supportive environment where adolescents can express and understand their emotions.

By using text-based interaction and natural language processing techniques, MindCapsule identifies emotional states from user input and provides personalized psychological guidance, including positive psychological suggestions and breathing exercises.

The practical goal is to develop an interactive emotional support system that encourages teenagers to recognize, accept, and express their emotions.

2. Development Timeline

June 14 - June 21: Project concept development and programming language selection.
June 21 - June 27: Program development, debugging, and demonstration video production.

During development, the project addressed common situations among students, such as feeling embarrassed about negative emotions, feeling uncomfortable receiving emotional support, or hiding disappointment after academic setbacks.

3. Project Background and Learning Outcomes

Development Tool: Visual Studio Code (VS Code).

Cognitive Change: Encouraging users to move from feeling ashamed of emotions toward accepting emotions as a normal human experience.

Skill Development: Improving the ability to express emotions, identify emotional states, analyze possible causes, and choose appropriate coping strategies.

4. Interdisciplinary Integration

The project combines programming technology, human-computer interaction, and positive psychology. Through a familiar text-chat interaction format, the system encourages adolescents to communicate their feelings and receive constructive emotional support.

5. Technical Components

The system includes a main program responsible for user interaction and emotional support generation, as well as a data module storing recommendation and encouragement messages.

6. Project Outcome

The final system demonstrates an AI-assisted emotional support platform that combines natural language processing with psychological principles to provide private and personalized emotional guidance.